

Solve all the questions:

1. Guess the limit of the sequence $\{x_n\}$, where $x_n = \frac{2n^2+1}{n^2+3n}$, $\forall n \in \mathbb{N}$ and verify its truth by ϵ definition on limit of a sequence. [5 Marks]

2. Find the conditions for $c > 0$, when the sequence $\{x_n\}$ defined by $x_1 = c$, $x_{n+1} = \frac{6(1+x_n)}{7+x_n}$ is monotone increasing and monotone decreasing. Check the convergence of the sequence $\{x_n\}$ and find its limit. [6 Marks]

3. (i) Find the upper and lower limit of the sequence $\{u_n\}$ where $u_1 = 1$, $u_{n+1} = \frac{(-1)^n}{2} \left\{ |u_n| + \frac{2}{|u_n|} \right\}$, $\forall n > 1$. [3 Marks]

(ii) Find the limit of the sequence $\{x_n\}$ where $x_n = \frac{1}{n} \{(2n+1)(2n+2) \dots (2n+n)\}^{\frac{1}{n}}$. [4 Marks]

4. (i) Let $b_n = 2 - \frac{(-1)^n}{n}$ and $a_n = b_n - b_{n+1}$, $\forall n \in \mathbb{N}$, then discuss the convergence/divergence of the series $\sum_{n=1}^{\infty} a_n$. [3 Marks]

(ii) Find the sum of the following series

$$\left(\frac{1}{2.3} + \frac{1}{2^2.3} \right) + \left(\frac{1}{2^2.3^2} + \frac{1}{2^3.3^2} \right) + \left(\frac{1}{2^3.3^3} + \frac{1}{2^4.3^3} \right) + \dots$$
 [4 Marks]

5. (i) If $\sum_{n=1}^{\infty} a_n$ converges, then find the limit of the sequence $\left\{ \frac{a_n-1}{n^n} \right\}_{n \in \mathbb{N}}$. [2 Marks]

(ii) Investigate the convergence and divergence of the following series

$$\frac{1^2}{2^2} + \frac{1^2.3^2}{2^2.4^2}x + \frac{1^2.3^2.5^2}{2^2.4^2.6^2}x^2 + \frac{1^2.3^2.5^2.7^2}{2^2.4^2.6^2.8^2}x^3 + \dots, \text{ for } x \in (0, \infty) \setminus \{1\}.$$
 [4 Marks]

6. Determine whether the following series is absolutely convergent or conditionally convergent

$$\frac{1}{1+a} - \frac{2}{8+a} + \frac{3}{27+a} - \frac{4}{64+a} + \dots \text{ where } a > 0.$$
 [5 Marks]

7. (i) Use sequential criterion for limit to evaluate the following limit

$$\lim_{x \rightarrow \infty} x^2 \operatorname{sgn}(\cos x), \text{ where } \operatorname{sgn}(x) \text{ is the sign or signum function.}$$
 [4 Marks]

(ii) A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} x^2, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational,} \end{cases}$$

Discuss the differentiability of $f(x)$ at $x = 0$.

[4 Marks]

8. (i) Prove that between any two real roots of the equation $e^x \sin x + 1 = 0$, there is at least one real root of the equation $\tan x + 1 = 0$.

[4 Marks]

- (ii) Is Mean Value Theorem valid for $f(x) = x^2 + 3x + 2$ in $[1, 2]$? Find the point $c \in (1, 2)$ at which the tangent to the curve $f(x) = 0$ is parallel to the straight line joining the points $(1, 6)$ and $(2, 12)$.

[2 Marks]

*****Best of Luck*****